

PALETTE-Aug: A Simple Texture-Preserving Augmentation for Medical Imaging Domain Generalization

Anonymous CVPR submission

Paper ID *****

Abstract

Introduction. Deep learning for medical image segmentation generalizes poorly across imaging modalities, and annotating data for every new domain is prohibitively costly. Synthesis-based augmentation methods such as SynthSeg address this by generating training images from label maps alone, but they require dense label maps and discard the source image’s texture. We introduce PALETTE-Aug, a texture-preserving alternative, and ask when texture matters for generalization. **Methods.** We benchmark PALETTE-Aug against five methods on eight segmentation tasks: five whose targets are bounded by an anatomical interface (healthy brain, multi-organ, healthy spine, mandible, pelvis) and three defined by tissue appearance (brain tumor, multiple sclerosis lesions, breast lesions). Each method trains on one modality and is evaluated in-domain and out-of-domain. An ablation isolates the effect of texture preservation. **Results.** PALETTE-Aug achieves the highest Dice on six of the eight tasks and significantly outperforms every competing method across them. In a six-task ablation, replacing noise fill with real texture raises per-task out-of-domain Dice by a relative 15% on brain tumor and 16% on multiple sclerosis lesions, both appearance-defined, but by at most 4% on interface-bounded tasks (healthy brain: −3%). Breast lesions, also appearance-defined, show no Dice gain. **Conclusion.** Texture preservation helps most on targets defined by tissue appearance rather than an anatomical interface, though not on all of them. PALETTE-Aug generalizes best overall across eight segmentation tasks.

1. Introduction

Medical image segmentation models are trained on one acquisition setting and are frequently applied to another: a different scanner, sequence, or imaging modality than the one seen during training. MRI contrast is a protocol choice, not a physical constant, so the same anatomy has different intensities across sequences. A network trained only on T1-

weighted brain images can fail outright on T2-weighted or FLAIR images [3]. Acquiring and annotating training data for every contrast a model might encounter is not practical: manual segmentation is expensive, and clinical sites rarely share raw data across institutions [33, 51]. The setting that matters for deployment is therefore single-source domain generalization: train on one labeled dataset in one contrast, then generalize to unseen contrasts and modalities without target-domain data [36, 45, 55, 57].

The dominant single-source strategy is label-generative domain randomization, popularized by SynthSeg [3]. It resamples intensities from a per-label Gaussian mixture over a dense anatomical label map, a voxel-wise parcellation of the whole image rather than the sparse annotation a segmentation dataset normally provides. The resulting images have arbitrary, unrealistic contrast, forcing the network to rely on shape and layout [3]. This is effective, but carries two structural costs. First, it is *label-driven*: the dense parcellation it needs is richer than the task’s own annotation, so it reintroduces the expensive labeling it aims to avoid. Second, and less discussed, it *does not preserve texture*: replacing every labeled region with piecewise Gaussian noise discards the real gradients, partial-volume boundaries, and unlabeled fine structure the acquired scan contains. Layout survives; texture does not.

We ask when keeping texture, rather than discarding it, helps contrast robustness. Our central finding is that the answer depends on the *task*: preserving real intensity structure helps most where the target is defined by tissue appearance, and much less where an anatomical interface separates it from its surroundings. Prior work shows that texture carries diagnostic signal in appearance-defined tasks: texture features separate MS lesions from normal-appearing white matter [56], and diffuse glioma infiltrates beyond every margin visible on conventional MRI [17, 47].

PALETTE-Aug, our single-source augmentation, keeps this texture and needs no dense label map. PALETTE-Aug groups the foreground intensities of a scan into classes with 1-D k -means, splits each class into spatial sub-regions with a Voronoi partition, and applies a signed random affine

remap to the *real* intensities of each sub-region. Edges, gradients and partial-volume detail therefore survive inside each sub-region, because existing intensities are remapped rather than replaced by noise. Only the contrast between regions is randomized, including sign inversions that no acquisition protocol produces. An optional last step uses labels, and only the sparse task labels that any segmentation dataset already has.

We test the mechanism by intervention. Keeping the partition and the region target means fixed, we replace SynthSeg-style noise fill with the real-intensity remap in fourteen training settings and measure the Dice change relative to the noise-fill Dice. Pooled per task, the gain is largest on the appearance-defined targets, brain tumor (+15%) and MS lesions (+16%), against at most 4% on interface-bounded tasks. The converse does not hold: several appearance-defined settings, breast lesions among them, show no Dice gain, so we state the result in one direction only (§4.3). A segmentation-free gradient measurement confirms that the swap changes texture: noise fill keeps only the partition edges, and real fill recovers most of the source texture.

We train one nnU-Net [22] configuration, varying only the epoch budget per dataset, and evaluate cross-contrast, cross-modality and cross-dataset transfer on eight tasks: healthy brain, MS lesion, brain tumor, abdominal organ, mandible, breast lesion, spine and pelvis. We compare against nnU-Net with its default augmentation, two variants of SynthSeg (the original and our re-implementation of its EM label-subdivision extension), SRCSM [48], and Auglab [34], a published cross-modality augmentation pipeline. PALETTE-Aug is the best method overall: it has the best Dice on six of the eight tasks and is significantly better than every competitor across tasks, on both Dice and boundary distance.

In summary, we make the following contributions:

- **When texture preservation matters, tested by intervention.** Replacing noise fill with the real-intensity remap, with the partition fixed, gives the largest gains relative to the starting Dice on appearance-defined targets, though not on every one. A segmentation-free gradient measurement confirms that the swap changes texture.
- **PALETTE-Aug**, a single-source contrast augmentation that needs no dense label map, only the task’s own labels. In every comparison it runs as one added operator inside the Auglab pipeline; the ablation also tests it alone.
- **Breadth evidence.** Across cross-contrast, cross-modality and cross-dataset transfer on eight tasks and sixteen datasets, with one configuration reused throughout, PALETTE-Aug is best on six tasks and significantly better than every competitor across tasks; where it does not win, we test whether the leader is truly ahead.

2. Related Work

2.1. Domain generalization for medical segmentation

Domain generalization (DG) targets unseen distributions without access to target data, unlike unsupervised domain adaptation, which assumes unlabeled target images and solves an easier problem [44]. Most medical DG methods are *multi-source*, interpolating across several labeled domains by meta-learning, feature alignment, frequency-space exchange, multi-frequency architectures, foundation-model scale, or diffusion-based stylization [2, 30, 33, 35, 42, 51] – all of which still rely on source diversity. We instead work in the *single-source* regime (one labeled dataset, no target data, no second source), which is routine in clinical deployment [36, 57]. This rules out cross-domain interpolation and makes architecture-coupled methods orthogonal to our contribution: our mechanism acts at the input level and leaves the segmentation network a fixed nnU-Net [22].

2.2. Synthesis-based contrast-agnostic training

The dominant single-source strategy is label-generative domain randomization, introduced by SynthSeg [3]: intensities are sampled from a per-label Gaussian mixture over an anatomical segmentation, yielding images of arbitrary contrast that force contrast-invariance. This family has the two costs noted in Sec. 1: it needs a dense anatomical label map per subject, and it replaces real texture with piecewise noise. SIAM [49] trades label *quantity* for *quality* with curated templates, but still needs a full dense parcellation. The lineage has been specialized to white-matter and MS lesions and to atlas-based expectation-maximization [6, 7, 28]. Among SynthSeg’s own variants we take the original (*noEM*), which synthesizes from the label map alone and never looks at the acquired scan, and the *EM* extension, which first splits every label, background included, into intensity-coherent sub-regions by expectation-maximization clustering of the real image, so that sparse task labels yield a richer map to synthesize from. SynthSeg introduced the EM extension for cardiac MRI [3, §5.4]; we re-implement it.

2.3. Texture as a segmentation signal

Independent work establishes that local texture, not only intensity level, carries information a segmentation model can use: second-order gray-level statistics discriminate image content beyond simple intensity [18], and quantitative texture features from tumor regions carry prognostic signal across CT, MRI and PET [27]. Closer to our setting, Kang et al. disentangle structure from texture in cross-domain image translation with a texture co-occurrence loss, targeting the same failure mode we do — naive translation distorts

fine texture and degrades downstream segmentation [24]; a texture-fidelity loss has likewise been added to GAN-based lesion augmentation [15]. For diffuse white-matter lesions, it has been argued directly that the lack of a sharp edge defeats boundary-based methods and favors texture-based segmentation [26]. None of this prior work, to our knowledge, connects texture preservation in an *augmentation* to which segmentation targets it helps; that connection, tested causally, is our contribution.

2.4. Single-source intensity and appearance augmentation

Where the first line of single-source methods synthesizes from a label map, a second manufactures appearance diversity directly from the real image, with no generative label model. General-purpose libraries stack intensity, spatial and artifact transforms on an nnU-Net trainer; the recent Auglab pipeline [22, 34] is one such GPU-accelerated framework and is the strongest competitor across our tasks, and BigAug stacks a comparable set of coarse intensity, quality and spatial perturbations under the same single-source DG protocol we use [57]. Our method inserts the PALETTE-Aug transform as one additional operator inside the Auglab pipeline, so the comparison against Auglab directly isolates PALETTE-Aug’s contribution. Closer to our operator are random-filter methods, which vary appearance with an untrained nonlinear filter: GIN applies randomly re-initialized convolutions for label-free appearance variants [36]; related variants establish the random-filter principle for natural images, perturb intensities per class using the task mask, add learned diffusion generators, or perturb the Fourier amplitude spectrum instead of the image itself [8, 31, 39, 45, 55, 58]. Our operator instead partitions the foreground into intensity classes and Voronoi sub-regions and applies a signed *affine* remap of the real intensities within each, invertible and gradient-preserving by construction. The closest peer is SRCSM [48]. Like PALETTE-Aug it perturbs region-wise in 3-D with one label-consuming step: it applies a fresh untrained convolution per anatomical class, smooths the resulting boundary artifacts, and histogram-matches test images to the source. They differ in the transform: SRCSM applies a random nonlinear convolution, PALETTE-Aug an affine remap that keeps the relative intensity structure inside each region.

3. Method

We describe PALETTE-Aug (§3.1), how it is applied during training (§3.2), and the causal ablation used to isolate texture preservation as the source of its effect (§3.3).

3.1. The PALETTE-Aug transform

Let x be a training volume min-max normalized to $[0, 1]$, and F its foreground voxels ($x_v > 0.01$). PALETTE-Aug

applies four steps at every training iteration (Fig. 1).

1. Intensity partition. We run 1-D k -means on the foreground intensities $\{x_v : v \in F\}$ with k sampled per iteration from $\{2, \dots, 6\}$, producing k intensity classes $C_1, \dots, C_k$ (Fig. 1, panel (b)). This partition groups voxels by intensity alone, with no label map, so it applies unchanged to any modality or contrast. Occasionally it is skipped and step 3 applies one global remap to the whole foreground; all step probabilities are in Sec. 7.

2. Spatial sub-parcellation. A Voronoi partition seeded with points sampled within each class C_i splits it into sub-regions $R_{i,1}, \dots, R_{i,m_i}$ that are spatially localized and similar in intensity (Fig. 1c). Two voxels of equal intensity in different parts of the anatomy can thus fall in different sub-regions and receive independent contrast changes. Each class is usually, but not always, sub-split this way (Sec. 7).

3. Signed affine remap. For each sub-region R , let $\bar{x}_R$ denote its mean intensity, $\mu_R \sim U(0, 1)$ a resampled target mean, and draw $\alpha_R \sim \pm U(0.5, 2.0)$ (a magnitude in $[0.5, 2.0]$ with an independently sampled sign). Every voxel $v \in R$ is remapped as

$$y_v = \mu_R + \alpha_R (x_v - \bar{x}_R). \quad (1)$$

Because Eq. (1) is a per-region affine function of the *real* intensity x_v , it preserves the relative structure of the signal within each region: edges, gradients, and partial-volume transitions between adjacent voxels are scaled, not replaced. Only the contrast between regions is randomized; $\alpha_R < 0$ also inverts local contrast, which no acquisition protocol produces. Background voxels are zeroed at this step (and remain zero through the rest of the pipeline). Half the time, the result is then Gaussian-blurred (Sec. 7).

4. Anatomical-label affine remap (optional). The unsupervised partition of steps 1–3 need not follow anatomy: two adjacent tissues of similar intensity can, by chance, receive the same target contrast, erasing their boundary. Step 4 corrects this: for each anatomical label ℓ , we draw a fresh (μ_ℓ, α_ℓ) and apply Eq. (1) *again* to ℓ ’s voxels of the step-3 output (Fig. 1d), so the label remap composes with the unsupervised one and keeps its internal structure. Since labels are independent of the unsupervised partition, the label remap restores a full-strength contrast step at every true label border, whether or not step 3 kept it; the anatomy itself is unchanged. It uses only the task’s own labels and is optional: without it, PALETTE-Aug reduces to steps 1–3, e.g. on unlabeled data. An optional second blur and a foreground z-score normalization then give the training input (Fig. 1e).

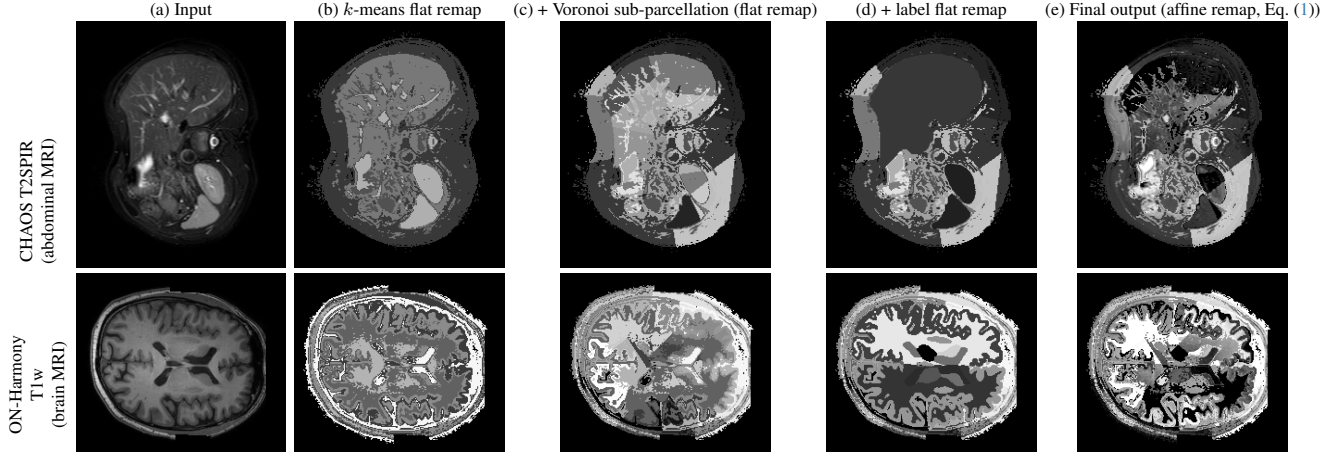


Figure 1. The PALETTE-Aug pipeline on one real CHAOS T2SPIR volume (top) and one real ON-Harmony T1w volume (bottom); every panel is a deterministic function of the same real volume (no external model), min-max normalized to $[0, 1]$ for display only. Panels (b)–(d) paint each region, sub-region or label flat with its target mean μ (Eq. (1)) to show the partition alone; panel (e), the actual training input, applies the full signed affine remap, keeping real texture, at region and, where available, label level.

3.2. Training protocol

PALETTE-Aug is applied on-the-fly as one contrast-augmentation operator inside an existing GPU-accelerated nnU-Net augmentation pipeline [22, 34], alongside that pipeline’s standard spatial and artifact augmentations; it adds no learned parameters and runs on the GPU inside that pipeline. One configuration (k -range, sub-region count, blur schedule, α range; Sec. 7) is set once and reused unchanged for every dataset and modality; no dataset gets its own hyperparameter sweep.

3.3. Causal ablation: isolating texture preservation

PALETTE-Aug differs from SynthSeg in two ways at once: the partition (unsupervised k -means and Voronoi vs. an anatomical label map) and the fill (remapped real intensities vs. Gaussian noise). A direct comparison cannot separate them.

We construct a controlled variant, *noise-fill*, with the same k -means/Voronoi partition and the same random target mean μ_R for each sub-region as Eq. (1). Instead of remapping the real intensities, it fills each sub-region with SynthSeg-style Gaussian noise, $\mu_R + \sigma_R \epsilon$ with $\epsilon \sim \mathcal{N}(0, 1)$ i.i.d. per voxel and σ_R drawn uniformly from $[0.05, 0.25]$ per sub-region; the label step is noise-filled the same way. The partition and the target means are PALETTE-Aug’s own, so a difference between the two isolates whether the real intensity structure inside each region is kept or replaced by noise. We report this comparison separately for *interface-bounded* targets (a normal structure bounded by another tissue class) and *appearance-defined* targets (a pathology identified by abnormal signal within a tissue), assigned by anatomy; the texture hypothesis predicts a larger

gain on the latter.

4. Experiments

4.1. Setup

Protocol. Each model trains on one real source modality and is tested on every modality of the same task, the protocol Auglab uses for spine [34], here over eight tasks and with external cohorts wherever one exists. No target-domain data is seen in training. All methods share one nnU-Net [22] configuration and schedule and differ only in input augmentation; the epoch budget is fixed per dataset (Sec. 7). Our full method adds the PALETTE-Aug transform as one operator to the Auglab pipeline [34], so it and Auglab differ by exactly that transform. Each method is trained on three train/validation folds, and all three models are scored on the same held-out test set. Significance tests are paired, use the patient (for ON-Harmony, the session) as the unit, and are Holm-corrected [20] (Sec. 9).

Datasets. We evaluate on eight tasks drawn from sixteen datasets (full roster in Tab. 3): abdominal organs, healthy brain, brain tumor, MS lesions, mandible, breast lesions, spine and pelvic musculoskeletal anatomy, across MRI, CT and dental CBCT. Brain tumor and MS lesions use BraTS2024-GLI [10] and Open-MS [29]. Each (task, training modality) pair is one *setting*, eighteen in all, tested on every contrast of its task, the training contrast included; *out-of-domain* (OOD) results exclude the training contrast, also in external cohorts. External cohorts are pooled into their task as extra held-out test columns, not reported as separate tasks. CHAOS [25] transfers to the CT of AMOS [23] and SLIVER07 [19]. Mandible (ToothFairy2 [5], CBCT only)

transfers to the CT and MR of HaN-Seg [38] and the CT of PDDCA [40], so every held-out contrast but its own CBCT comes from another dataset. Breast (I-SPY2 [32]) transfers to Duke-Breast-MRI [13, 41], I-SPY1 [21] and ACRIN-6698 [37], whose diffusion-weighted MRI resembles neither training modality. Spine (a CT/Dixon cohort) transfers to the T1w and T2w of SPIDER [50]. Pelvis is built from the TotalSegmentator CT and MRI releases [9, 53], two unpaired cohorts. ON-Harmony [52] reference labels are silver-standard FreeSurfer [12] parcellations of the T1w, registered to the other contrasts. External volumes that image more of the body than the training cohort are cropped to its field of view before prediction, identically for every method (Sec. 8).

Baselines. We compare against five methods. (1) *Baseline*, nnU-Net with its default augmentation. (2) *SynthSeg-noEM* [3], which synthesizes each image from the label map alone and ignores the acquired scan; it receives each task’s training labels, which form a dense parcellation only on ON-Harmony. (3) *SynthSeg-EM*, our re-implementation of SynthSeg’s EM extension [3, §5.4], which first subdivides every label by expectation-maximization clustering of the image. (4) *SRCSM* [48], a semantic random-convolution augmentation outside both the label-generative and the Auglab families. (5) *Auglab* [34], a published GPU augmentation pipeline. It is a competitor in its own right, not an ablation of ours; since our method adds one operator to it, comparing the two also tests that operator.

4.2. Main comparison

PALETTE-Aug is the best method overall. PALETTE-Aug has the best Dice on six of the eight tasks. Across tasks it is significantly better than every competitor ($p < 10^{-10}$ each, Tab. 1). Within single tasks it significantly outperforms every competitor on five tasks (BraTS-GLI, CHAOS, Open-MS, Breast, Spine), three of five on Mandible and Pelvis, and two of five on ON-Harmony. The cross-task test generalizes to new patients on these eight tasks, not to new tasks. Counting each task as one unit instead, PALETTE-Aug beats Auglab on seven of eight tasks (one-sided sign test $p = 0.035$) and SRCSM on five of the seven tasks where SRCSM was run ($p = 0.23$).

Given sparse labels, SynthSeg-noEM paints one Gaussian background around the labeled structures and falls below the baseline on six of the eight tasks, every one with sparse labels; SynthSeg-EM, which subdivides the labels by clustering the image, closes most of that gap.

No competitor is reliably second. The best competitor alternates between Auglab (BraTS-GLI, Open-MS, Breast, Spine) and SRCSM (ON-Harmony, Mandible, Pelvis), the two tie on CHAOS, and SRCSM fails on Breast (16.4 Dice).

Picking the right one would require knowing the answer in advance. PALETTE-Aug is first or second on every task but one.

Where PALETTE-Aug does not win. The one-sided test above cannot show that a competitor is ahead, so on ON-Harmony and Mandible we also ran the reverse test on the same patient-level design (Holm-corrected across competitors within the task). On ON-Harmony SRCSM leads by 0.1 Dice and neither method is significantly ahead. On Mandible SRCSM (+1.3 Dice, $p = 6 \times 10^{-4}$) and Auglab (+0.6, $p = 2.5 \times 10^{-3}$) are significantly ahead, and PALETTE-Aug is third. In the Mandible ablation (§4.3), real fill helps significantly; Dice drops only once Auglab is added (Tab. 9).

Boundary distance. The 95th-percentile Hausdorff distance (HD95) is reported in the supplement (Tab. 7). PALETTE-Aug has the lowest HD95 on six of the eight tasks (BraTS-GLI, CHAOS, ON-Harmony, Open-MS, Spine, Pelvis) and is significantly better than every competitor across tasks. It ties Auglab on Breast and is significantly behind SRCSM, Auglab and the baseline on Mandible.

4.3. Causal ablation: when does texture matter?

Does keeping the real texture help beyond PALETTE-Aug’s partition, and on which targets? We test this with a ladder in which each rung adds one ingredient: baseline $\rightarrow$ $+k$ -means $\rightarrow$ $+label$ remap $\rightarrow$ $+Voronoi$ $\rightarrow$ real-intensity fill $\rightarrow$ $+Auglab$. The three partition rungs fill every region with SynthSeg-style Gaussian noise. All rungs share one pipeline (Sec. 3) and differ only in which steps are on; before $+Auglab$, the only other augmentation is spatial (with flips). This step from $+Voronoi$ to real fill, the *fill swap*, changes only the fill: partition, sub-region count, target-mean distribution and both blur passes are unchanged. Any change at this step therefore comes from keeping the real within-region intensity structure instead of noise.

In our data every large relative gain from real fill is on an appearance-defined target, but not every appearance-defined target shows one (Fig. 2 and Tab. 2). We report each gain relative to its noise-fill Dice, because the same few points are a small share of a high starting Dice and a large share of a low one (Sec. 5 notes the alternative); the three largest gains are the same relative or absolute. They are BraTS-GLI T1c (+43.6%, +13.3 Dice) and T1n (+18.1%, +7.2), both significant after correction over the fourteen rows, and Open-MS FLAIR (+39.3%, +7.7), which improves for all eight test patients but is not significant after that correction. No interface-bounded row gains more than 6.2%; the largest, CHAOS T2spir, adds 5.1 points from a noise-fill Dice of 82.0. Pooled per task (Fig. 2), the step

	BraTS-GLI	CHAOS	ON-Harm.	Open-MS	Mandible	Breast	Spine	Pelvis	p vs ours
Baseline	23.4	29.5	35.0	21.0	59.2	35.0	48.5	29.8	$< 10^{-10}$
SynthSeg-noEM	8.8	60.9	64.8	4.3	53.6	16.5	47.5	15.0	$< 10^{-10}$
SynthSeg-EM	43.1	84.0	66.3	33.9	76.3	39.6	80.6	62.5	$< 10^{-10}$
SRCSM	41.5	84.6	66.6	33.0	79.0	16.4	—	65.3	$< 10^{-10}$
Auglab	44.8	84.6	66.1	34.3	78.3	40.1	80.9	64.3	$< 10^{-10}$
PALETTE-Aug (ours)	45.9	85.9	66.5	38.5	77.7	41.2	82.3	65.6	—
competitors beaten ($p < 0.05$)	5/5	5/5	2/5	5/5	3/5	5/5	4/4	3/5	

Table 1. Task-level Dice (%). Each column pools the task’s test cases of every contrast (the training contrast included), all source modalities and any external cohorts, weighting contrast groups equally so a modality with several external sources does not outvote one with fewer (Sec. 9). p : Holm-corrected one-sided sign-flip test on the macro-averaged per-patient difference, stratified by task with tasks weighted equally. We print no cross-task mean Dice: averaging whole-liver and breast-lesion Dice gives a number no clinician would recognize. The bottom row counts the competitors PALETTE-Aug beats within each task alone (same test). Spine has no SRCSM run (Tab. 3). HD95 is in Tab. 7.

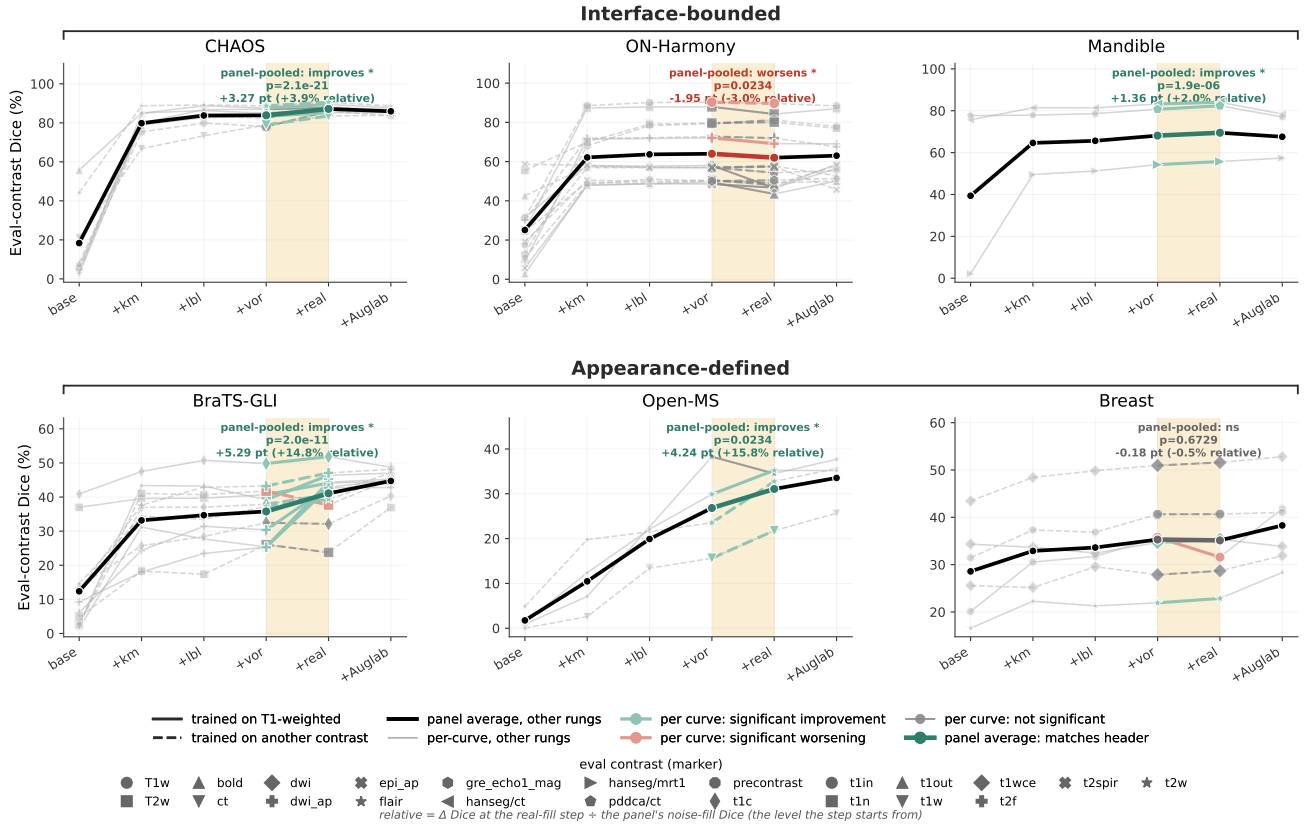


Figure 2. The one-variable test, one panel per task with a ladder, grouped by target type. Each thin line is one (training modality, held-out contrast) pair, external cohorts included; only its real-fill segment (shaded band) is colored, by that pair’s significance: teal improves, red worsens, grey is not significant. The bold black line is the panel average, colored by the panel test annotated above it: one paired Wilcoxon test on *patients* (each counted once across contrasts and models) pooling all training modalities, Holm-corrected across the six panels. rel. = panel gain / mean noise-fill Dice. Two of the three appearance-defined panels improve significantly and strongly (BraTS-GLI +14.8%, $p = 2.0 \times 10^{-11}$; Open-MS +15.8%, $p = 0.023$) and Breast does not move. The interface-bounded panels move much less in relative terms: CHAOS (+3.9%) and Mandible (+2.0%) gain, and ON-Harmony worsens (−3.0%; significant with its eight sessions as units, which come from four subjects). Solid/dashed = trained on a T1-weighted vs. other contrast; marker = held-out contrast. Per-row numbers in Tab. 2; HD95 in Fig. 4 and Tab. 8.

Task	noise	Δ	rel. (%)	p
<i>Interface-bounded</i>				
CHAOS T1in	85.7	+1.43	+1.7	4.0e-12
CHAOS T2spir	82.0	+5.11	+6.2	1.0e-20
ON-Harmony T1w	63.2	-5.18	-8.2	0.070*
ON-Harmony T2w	64.2	-1.55	-2.4	0.070*
ON-Harmony DWI	64.6	+0.78	+1.2	0.094*
Mandible CBCT [‡]	68.1	+1.36	+2.0	4.9e-6
<i>Appearance-defined</i>				
Open-MS FLAIR	19.6	+7.69	+39.3	0.070*
Open-MS T1w	34.1	+0.79	+2.3	1.0*
BraTS-GLI T1n	39.8	+7.22	+18.1	2.8e-11
BraTS-GLI T1c	30.5	+13.27	+43.6	4.3e-12
BraTS-GLI T2w	41.0	+0.87	+2.1	0.38
BraTS-GLI FLAIR	31.9	-0.19	-0.6	1.0
Breast T1-CE	30.8	-0.85	-2.8	0.26
Breast T2w	39.8	+0.50	+1.3	0.094

Table 2. Replacing SynthSeg-style noise fill with the real-intensity remap, with the partition and region target means unchanged, for every training modality of every task with a ladder. noise: out-of-domain Dice with noise fill; Δ : Dice change at the swap; rel.: Δ relative to the noise-fill Dice (bold: above 15%). p : two-sided paired Wilcoxon on patients, pooled over the row’s held-out contrasts, Holm-corrected across all fourteen rows. *With eight test patients (Open-MS) or eight sessions of four subjects, two sites each (ON-Harmony), no row reaches $p < 0.05$ after correction even when all eight move together: Open-MS FLAIR (8/8 improve), ON-Harmony T1w and T2w (8/8 worsen). [‡]Held-out contrasts are cross-dataset (HaN-Seg, PDDCA), since ToothFairy2 is CBCT only. HD95 in Tab. 8.

is worth +14.8% on BraTS-GLI and +15.8% on Open-MS, against +3.9% on CHAOS, +2.0% on Mandible and -3.0% on ON-Harmony. Not every appearance-defined setting gains: Open-MS T1w, BraTS-GLI T2w and FLAIR, and both Breast rows (-2.8% and +1.3%, four cohorts pooled by contrast) show no Dice gain.

ON-Harmony labels 31 brain regions in the FreeSurfer label set [12]: ventricles, cortex, white matter, cerebellum, brainstem and eight deep grey nuclei. Nearly all are bounded by another tissue class (CSF against tissue, grey against white matter), so we group it with CHAOS as interface-bounded. At the task level it is the only task where the swap significantly hurts, with sessions as units. It is also the only task labeling nearly the whole brain; with one such task, we cannot tell whether that is why.

Does the fill swap change texture? The ladder shows that the fill swap changes Dice, not that it changes texture. We check this without any segmentation, with the normalized-gradient-field (NGF) similarity [16] between each augmented volume and its source scan. NGF measures how well image gradients align: it is high when edges and

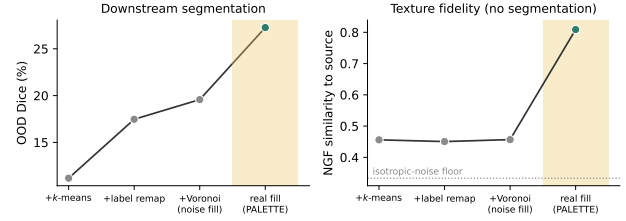


Figure 3. Noise versus real fill on Open-MS FLAIR. Left: out-of-domain Dice at each rung. Right: NGF similarity of the augmented volume to its source (mean $\pm$ s.e. over 30 scans), a segmentation-free measure of retained texture. The noise-fill rungs keep only the partition edges while Dice rises; the real-fill swap (shaded) lifts both.

fine structure are kept and falls to about 1/3 for isotropic noise. Figure 3 follows NGF and Dice along the same rungs on Open-MS FLAIR. The three noise-fill rungs stay at NGF 0.45–0.46: above the noise floor, since partition edges survive, but far below the source. Their Dice still rises from 11.2 to 19.6, through contrast randomization, not texture. Only the real-fill swap raises both, NGF to 0.81 and Dice to 27.3. On the same scale, PALETTE-Aug without Auglab (the real-fill rung) keeps significantly more source texture than Auglab (paired Wilcoxon, 30 scans; FLAIR $p = 3.2 \times 10^{-3}$, T1w $p = 1.2 \times 10^{-6}$), and the SynthSeg variants stay far below the source (Tab. 14).

Supporting analyses. The supplement adds a component ablation (Sec. 11), per-setting results, every ladder rung and the HD95 ablation (Sec. 9).

5. Discussion and Limitations

Why anatomy predicts the largest effects. Liver, spleen and kidney are encapsulated organs with a clear physical boundary, and edge-following methods have long segmented them well [46]. An augmentation that scrambles their internal texture but keeps the boundary loses little. Diffuse glioma has no such boundary: tumor cells infiltrate beyond both the enhancing margin and the FLAIR abnormality [17, 47], so the target must be inferred from appearance. MS lesions, though typically well demarcated, are identified by abnormal signal within white matter, and texture features separate them from normal-appearing white matter [56]. This account interprets the intervention of Sec. 4.3; it is not independent evidence.

Not every appearance-defined target gains. Open-MS T1w, BraTS-GLI T2w and FLAIR, and both Breast rows show no Dice gain. We have two hypotheses. First, the effect depends sharply on the training contrast (BraTS-GLI: +13.3 from T1c, +7.2 from T1n, about zero from T2w

and FLAIR), so whether its texture transfers may matter as much as the target; with T2w training, real fill helps some labels and hurts others (Sec. 9). Second, Breast pools four cohorts from different trials, which may dilute a cohort-specific effect. On interface-bounded tasks, where texture adds little relative to the starting Dice, most of the gain arrives with the texture-free partition rungs (Tab. 9), that is, from the wide range of region-to-region contrasts PALETTE-Aug samples.

Limitations. We assign each target to a group by its anatomy, not by a measured boundary statistic. Target type is confounded with difficulty: every interface-bounded row starts above 63% noise-fill Dice and every appearance-defined row below 42%, so dividing by the starting Dice favors the latter; as a share of the remaining error, CHAOS T2spir would rank first. The between-group comparison is descriptive: with three ladder tasks per group, no test of it can reach $p < 0.05$. Open-MS has eight test patients and ON-Harmony eight sessions of four subjects, so their single-setting effects can be consistent without reaching significance, and ON-Harmony’s session-level tests are optimistic. Spine was run on a separate pipeline and has no SRCSM arm. The cross-task test covers new patients, not new tasks; with the task as the unit, the advantage over SRCSM is not significant. PALETTE-Aug targets the single-source setting and is complementary to multi-source domain generalization, which assumes several labeled source domains.

6. Conclusion

We asked when preserving real image texture matters for contrast-robust segmentation. Relative to the noise-fill Dice, preserving texture helps most where the target is defined by tissue appearance rather than an anatomical interface: +15% on brain tumor and +16% on MS lesions per task, against at most 4% on interface-bounded tasks. Not every appearance-defined target gains. PALETTE-Aug, which preserves texture and needs no dense label map, is the best method overall across eight tasks and sixteen datasets. The task-level account makes a testable prediction for new tasks: a large relative gain from texture preservation should not be expected where the target has an anatomical interface.

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PALETTE-Aug: A Simple Texture-Preserving Augmentation for Medical Imaging Domain Generalization

Supplementary Material

7. Training configuration

All methods share one nnU-Net [22] configuration; only the input augmentation differs. Nothing below is tuned per dataset.

Training schedule. 3D full-resolution nnU-Net, SGD with Nesterov momentum 0.99, initial learning rate 10^{-2} with the nnU-Net `polyLR` decay, 250 mini-batches per epoch, combined Dice and cross-entropy loss, deep supervision on. Patch size, spacing and batch size are whatever the nnU-Net planner selects for that dataset, identically for every method. The epoch budget is fixed per dataset and shared by all methods on it: 2500 (BraTS-GLI), 2000 (ON-Harmony, Open-MS), 1000 (ToothFairy2, I-SPY2) and 200 (CHAOS, Pelvis); Spine was trained on a separate pipeline. Each method is trained as three models on three train/validation folds, all scored on the same held-out test set; folds are the same across methods.

PALETTE-Aug hyper-parameters. Set once and never tuned per dataset: k -means class count drawn per sample from $\{2, \dots, 6\}$, skipped in favor of a single global remap with probability 0.10; Voronoi sub-parcellation of each intensity class with probability 0.60 (else left as one region); signed affine coefficient $\alpha \in \pm U(0.5, 2.0)$ applied per sub-region as in Eq. (1) (always applied, to whatever partition results from the k -means/Voronoi steps); per-label affine decoupling applied independently per label with probability 0.5 (skipped for any label with under 4 foreground voxels in that sample); Gaussian blur σ drawn from $\{0, 0, 0, 0.3, 0.5, 0.8\}$, applied twice (after the region-level and after the label-level remap), so applied half the time at each point. The transform is applied on the GPU, on-the-fly, to a fixed 50% of training iterations (*train050*); the remaining half of iterations use the un-transformed real image.

Validation-time configuration. *val000* (the configuration reported in the main paper) applies no synthetic contrast to the validation split, matching every competing method and keeping model selection like-for-like. *val100* applies PALETTE-Aug to the entire validation split; it is reported in Sec. 10.

Baseline configurations. Both SynthSeg variants run through Auglab's GPU implementation of the SynthSeg

generator, with uniform intensity priors and with the generator's own spatial and flip steps disabled (the pipeline's spatial augmentation is shared by all methods). They receive each task's training label map; SynthSeg-EM additionally subdivides it on the fly by expectation-maximization clustering of the image (2 sub-regions per foreground label, 3–10 for the background, 20 iterations), where the original work precomputes these maps [3, §5.4]. Each sub-region then receives a randomly sampled intensity, as in SynthSeg. Both SynthSeg variants and SRCSM synthesize on every training iteration because their original papers train that way, so we keep that setting when running them; PALETTE-Aug synthesizes on 50% of iterations. SRCSM is a training-time augmentation only: no test-time intensity matching is applied, and every method is predicted with the same nnU-Net predictor, without test-time augmentation. SRCSM and Auglab otherwise use their published default configurations. No competing method was re-tuned in our favor, and all use the same folds, schedule and epoch budget as above.

Compute. Training was performed on NVIDIA L40S and H100 GPUs. A single fold costs from under an hour (CHAOS, 200 epochs) to roughly three days (BraTS-GLI, 2500 epochs); the full study is dominated by the six methods $\times$ eighteen settings $\times$ three folds of the main comparison, plus the fourteen ablation ladders of Sec. 4.3.

8. Datasets and evaluation protocol

All evaluation is performed on held-out test cases only. For each task the model is trained on a single source contrast and evaluated on every contrast of that task's test split, including the training contrast itself; training and validation subjects never appear in any reported number.

ON-Harmony [52] — 31-region healthy-brain segmentation. Reference labels are *silver standard*: FreeSurfer parcellations [12] of each session's T1w, grouped into 31 regions and rigidly registered with ANTs into the native space of the same session's other contrasts. They are not manual annotations, so absolute Dice here is agreement with an automated reference; every method is scored against the same reference. The test set is eight sessions of four subjects, each scanned at two sites.

Task	Dataset	Train modalities	Test modalities
Multi-Organ	CHAOS	T1-in, T2-SPiR	T1-in, T1-out, T2-SPiR, CT
	AMOS	—	CT, MRI
	SLIVER07	—	CT
Brain Glioma	BraTS2024-GLI	T1n, T1c, T2w, FLAIR	T1n, T1c, T2w, FLAIR
Brain MS	Open-MS	FLAIR, T1w	FLAIR, T1w, T2w
Healthy Brain	ON-Harmony	T1w, T2w, DWI	T1w, T2w, BOLD, DWI, EPI, GRE
Mandible	ToothFairy2	CBCT	CBCT
	HaN-Seg	—	CT, MR-T1
	PDDCA	—	CT
Breast	I-SPY2	T1-CE, T2w	T1-CE, T2w
	Duke-Breast-MRI	—	T1-CE, pre-contrast
	I-SPY1	—	T1-CE, pre-contrast
	ACRIN-6698	—	DWI
Spine [§]	CT/Dixon cohort	CT, Dixon in	CT, Dixon fat/in/water
	SPIDER	—	T1w, T2w
Pelvis	TotalSegmentator	CT, MRI	CT, MRI

Table 3. Per-task dataset roster. Rows marked “—” are cross-dataset transfer targets only, never trained on; their contrasts are extra held-out test columns of the task they sit under, not tasks of their own. Mandible’s non-CBCT test contrasts all come from other datasets, since ToothFairy2 acquires CBCT alone. Pelvis’s CT and MRI come from different patients. [§]Spine results were computed on a separate pipeline with the same method suite except SRCSM (Sec. 8). Counts are in Tab. 4.

Open-MS [29] — MS-lesion segmentation with manual consensus annotations, evaluated across FLAIR, T1w and T2w.

BraTS2024-GLI [10] — post-treatment glioma sub-region segmentation (necrotic core, surrounding FLAIR hyperintensity, enhancing tumor, resection cavity), trained on each of T1n, T1c, T2w and T2-FLAIR and evaluated on all four.

CHAOS [25] — abdominal organ segmentation (liver, spleen, both kidneys), evaluated on the CHAOS MR contrasts (four held-out MR patients) and CT (20 held-out patients), and on two external cohorts: CT and T2w MRI from AMOS [23] and liver-only CT from SLIVER07 [19]. In the pooled CT group, liver comes from CHAOS, AMOS and SLIVER07 and spleen and kidneys from AMOS only (CHAOS CT is annotated for the liver only). CHAOS images a restricted axial field of view, so the full-torso external volumes are cropped to the CHAOS-equivalent axial slab (median CHAOS margins around the kidneys for AMOS and the liver for SLIVER07) before prediction; the network sees, and is scored on, only the cropped volume.

ToothFairy2, HaN-Seg and PDDCA [5, 38, 40] — mandible segmentation, trained on dental cone-beam CT, the one non-CT, non-MRI training modality here. ToothFairy2’s annotation coverage is cohort-dependent, so every column of this task is scored on the mandible alone (teeth excluded, matching the external references), which keeps

the numbers being averaged commensurable. ToothFairy2 acquires CBCT only, so apart from its own 71 CBCT test cases its held-out contrasts all come from other datasets: HaN-Seg CT and MR-T1 (42 cases each) and PDDCA CT (40 cases), the two CT sources pooled as one contrast group. ToothFairy2’s volumes are stored superior-inferior inverted relative to the external cohorts — an inversion invisible to the image header, found by checking anatomy directly — so the external test volumes are flipped into ToothFairy2’s convention before inference, which leaves Dice and HD95 unchanged under the rigid transform. The correction matters for the comparison: a method without flip augmentation (SRCSM) was penalised by the inversion far more than the others, and it is after this correction that SRCSM leads this task.

Breast: I-SPY2, Duke-Breast-MRI, I-SPY1, ACRIN-6698 [13, 21, 32, 37, 41] — breast-lesion segmentation on malignant-only cohorts. I-SPY2 (CC BY 4.0) trains, on its first-post-contrast T1 (T1-CE) and its T2w; its reference is the trial’s functional tumor volume mask. Three external cohorts are test-only. Duke-Breast-MRI and I-SPY1 come through the MAMA-MIA release, whose expert primary-lesion protocol both share, and contribute T1-CE and pre-contrast columns; Duke-Breast-MRI is CC BY-NC 4.0, a non-commercial term no other cohort here carries. ACRIN-6698, the diffusion sub-study of I-SPY2 with no patient in common with our I-SPY2 subjects, contributes the baseline diffusion-weighted acquisition with the trial’s whole-tumor DWI region, a contrast neither training modality resembles. Because I-SPY2 is mostly acquired unilaterally, bilateral external volumes are cropped to the lesion side and, for the full-thorax acquisitions, also anterior-posteriorly to the breast-plus-chest-wall extent of I-SPY2’s own crops; without the second crop every method’s HD95 is dominated by the thorax. I-SPY2’s DCE series are acquired unilaterally at some sites and bilaterally at others; both field-of-view variants are kept per patient, and the significance tests treat them as one patient.

Field-of-view matching. A model tested on a field of view it never saw in training can fail for reasons that have nothing to do with contrast or modality. Where an external cohort images more of the body than the training cohort, we therefore crop its volumes before prediction, so that the shift left between training and test is, as far as possible, the contrast or modality alone. CHAOS (AMOS, SLIVER07): the axial slab CHAOS covers, placed with the median CHAOS margins around the kidneys (AMOS) or the liver (SLIVER07). Mandible (HaN-Seg, PDDCA): a $144 \times 128 \times 80$ mm box centred on the mandible, close to the 90th percentile of ToothFairy2’s training extent, enlarged (at most to $154 \times 154 \times 89$ mm) only when needed

to contain the whole mandible. Breast (Duke-Breast-MRI, ACRIN-6698): the lesion-side half of bilateral volumes, then, for full-thorax acquisitions, an anterior-posterior window from 27 mm in front of the skin to 147 mm behind it, the measured geometry of I-SPY2’s unilateral acquisitions, extended if needed to contain the lesion; I-SPY1 is natively unilateral and is not cropped. The network sees, and is scored on, only the cropped volume. Each crop is placed using the reference annotation (organ, mandible or lesion position). This reveals the target’s position, not its extent, and is identical for every method, so it does not favour any method; but absolute Dice and HD95 are better than a pipeline that had to localize the target first would obtain.

Spine [50] — vertebrae, intervertebral discs and spinal canal. Results for this task were computed on a separate pipeline with the same method suite except SRCSM, trained on CT and Dixon in-phase MRI of a CT/Dixon cohort and tested on that cohort’s held-out CT and Dixon fat, in-phase and water images and on the T1w and T2w of SPIDER. Dice and HD95 are computed on each whole structure per label (the pipeline’s `global_bin` metrics). No SRCSM arm was run, so Spine is absent from every SRCSM comparison.

Pelvis [9, 53] — eleven pelvic and hip musculoskeletal structures (both hips, sacrum, and the gluteus maximus, medius and minimus and iliopsoas on each side), from the TotalSegmentator CT (CC BY 4.0) and MRI (CC BY-NC-SA 2.0) releases. The two are unpaired cohorts, and the MRI release mixes sequences by design, so its MRI column is a sequence-heterogeneous test rather than one named contrast. CT is subsampled to 200 cases to keep training time comparable with the other tasks.

9. Per-setting results

Tab. 1 in the main paper reports results at the task level, pooling each task’s held-out test contrasts, all of its source modalities and its external cohorts. This section reports the same evaluation broken out into the eighteen individual source-modality settings. Every number is a cross-fold, cross-class average over all held-out contrasts of that setting, under the protocol of Sec. 4.1.

Statistical test construction. All tests are paired, and the unit of independence is the *patient*, never the fold, the image or the contrast. A patient’s score is its mean over labels and evaluated folds; where the same patient is scored in several held-out contrasts, or by the models of several training modalities, those scores move together as one unit under the null. Treating them as separate samples would be pseudo-replication and would inflate the effective sample size — a

Task	Train	Test
CHAOS	16	24
ON-Harmony	84 ^a	8
Open-MS	22	8
BraTS2024-GLI	630	70
ToothFairy2	408	71
I-SPY2	476	84
Pelvis (CT / MRI)	170 / 156	30 / 27

Table 4. Per-task subject counts for the training cohorts, pooled across the 3 folds used throughout (± 1 between folds). ON-Harmony counts are scan sessions; the eight test sessions are four subjects, each scanned at two sites. External cohorts are test-only and counted separately: AMOS and SLIVER07 (CHAOS), HaN-Seg and PDDCA (Mandible), Duke-Breast-MRI (291), I-SPY1 (167, after excluding three bilateral cancers and one implant by a rule fixed before prediction) and ACRIN-6698 (371) for Breast, and SPIDER (171 T1w, 219 T2w) for Spine, whose own CT/Dixon test split has 27–29 cases per contrast. I-SPY2 counts are patients; one held-out patient can yield more than one T2w image (both field-of-view variants). Pelvis MRI keeps 183 of the 298 released cases. ^a76 for the T2w-trained models.

real risk here, since several tasks test a handful of patients under many contrasts (ON-Harmony: eight sessions of four subjects, six contrasts). For ON-Harmony the unit is the session, since its four subjects leave a test that could never reach significance; two sessions of one subject are not fully independent, so its *p*-values are, if anything, optimistic, and we do not rest any claim on ON-Harmony alone. Per-setting comparisons (below) use the same test within each setting; the ablation ladders use the paired Wilcoxon signed-rank test [54], since there each setting is judged on its own. All tests are Holm-corrected [20] for multiple comparisons.

Why the aggregate tests are sign-flip tests on a macro-average. The aggregate tests (Tab. 1 and the per-dataset headline blocks) are not paired Wilcoxon signed-rank tests, which is the more usual choice in this literature: SynthSeg+, from the lineage we compare against, marks a method as better than its competitors using a one-sided Bonferroni-corrected Wilcoxon signed-rank test on paired Dice [4]. We therefore give our reason rather than leave it implicit. The *estimand* we test, by contrast, is not unusual at all — it is the same equal-weight-per-stratum principle the Medical Segmentation Decathlon adopted, for the same reason, as set out below.

The estimand. What we claim is $\text{macro}\Delta$: the mean, over held-out contrasts, of that contrast’s mean paired per-case difference (ours minus competitor) — equal weight per contrast, which is exactly what the *all* column of every table reports, so the test and the table measure the same thing. Equal weighting is not cosmetic here. The claim under test

is cross-contrast generalization, and a contrast’s held-out case count reflects how large a public cohort happens to be, not how much that contrast should say about generalizing. CHAOS is the extreme case: its held-out CT evidence includes an external cohort of 100 patients, while each of its three MR contrasts has 4 (cases counted once per training modality: 200 against 8). Weighting by case count would let one CT cohort cast more than half the votes in a test whose conclusion is about transfer across MR contrasts.

This is the same problem, and the same answer, as the Medical Segmentation Decathlon’s. Facing sample sizes that “varied heavily across all tasks and target ROIs,” it ranked algorithms within each target ROI, averaged those ranks per task, and averaged the per-task ranks for the final score — choosing that over a flat average across ROIs precisely because the flat scheme means “tasks with multiple target ROIs ... would be over-weighted” [1]. Our macro Δ applies the same principle one level across, with the held-out contrast in place of the target ROI. Where we differ is only in how that weighted quantity is reached: the Decathlon arrives at it through counts of pairwise Wilcoxon outcomes at an uncorrected per-comparison α , whereas we test the weighted difference itself and Holm-correct across competitors.

Why not a pooled Wilcoxon. The signed-rank test is defined on a single undifferentiated sample of paired differences. Applying it to the concatenation of all contrasts therefore tests a case-weighted null — the imbalance above, reintroduced — and there is no standard reweighting of the signed-rank statistic that recovers the equal-per-contrast estimand. The disagreement this causes runs in both directions: a pooled test can declare a tenth of a Dice point significant because one large cohort supplies most of the cases, or miss a consistent effect confined to the smaller contrasts.

The test. Under the null that the two methods are exchangeable on a given patient, that patient’s paired differences are equally likely to carry either sign. Sign-flipping is therefore a valid randomization null for *any* statistic computed from those differences [11], macro Δ included, which is what makes it applicable where the signed-rank test is not. We report the one-sided p in the direction of the stated hypothesis, with the two-sided p retained as a secondary column, and give a 95% confidence interval from a bootstrap that resamples patients within groups of patients sharing the same set of contrasts, so each source keeps its observed size.

Evaluating the tail. The statistic is a weighted sum of per-patient contributions, so its sign-flip null distribution can be computed rather than sampled. We enumerate all 2^M sign patterns of the $M=20$ patients with the largest contributions exactly, and treat the remaining, lighter patients as a Gaussian term with their exact null variance; with twenty patients or fewer the test is therefore exact. A plain normal approximation to the whole null, which an earlier version of

this analysis used, is badly conservative in exactly the designs we have, where a few patients spanning several contrasts carry most of the null variance: on the CHAOS table it returned $p = 0.013$ for a comparison that a 200,000-draw Monte-Carlo test places below 5×10^{-6} . The enumerated tail is deterministic and reaches far tails without the $1/B$ floor of resampling. Its corollary is a hard floor for small cohorts: eight patients admit $2^8 = 256$ sign patterns, so no one-sided p can fall below $1/256$ however consistent the effect.

What is reported alongside. Every generated significance report also carries a per-contrast table of mean Δ with the raw, uncorrected Wilcoxon p for each contrast individually, and the ablation ladders use the paired Wilcoxon throughout. The standard test is reported; it is simply not the one the pooled headline numbers rest on.

Contrast-group pooling. For CHAOS, the same tested modality is sometimes evidenced by more than one source: CHAOS’s CT evidence comes from its own held-out CT split plus AMOS and SLIVER07. Averaging every raw column with equal weight — the natural first implementation — silently lets a modality with more contributing sources outvote one with fewer in both the displayed *all* average and the significance test, since it then supplies more of the equally-weighted columns being averaged. We instead pool disjoint sources of the *same* modality by concatenating their cases into one group before any averaging happens, then average modality groups with equal weight — for CHAOS this additionally means averaging equal-weight across liver/spleen/kidney *within* the pooled CT evidence, since AMOS’s own CT column already blends all three organs into one number before it reaches this step. The same pooling applies to Mandible (HaN-Seg and PDDCA both supply CT) and Breast (I-SPY2, Duke-Breast-MRI and I-SPY1 supply T1-CE, and Duke-Breast-MRI and I-SPY1 also supply pre-contrast); the remaining tasks have no contrast evidenced by more than one source.

Tables 5 and 6 break each task column of Tab. 1 into its training modalities. Where a margin is not significant in a single setting, the honest reading is that one setting cannot separate the methods, not that they are equivalent: several settings have few test units (eight patients on Open-MS, eight sessions on ON-Harmony), which is why the main claims rest on the task- and cross-task-level tests.

Boundary distance at task level. Tab. 7 gives the HD95 counterpart of Tab. 1. Across tasks PALETTE-Aug is significantly better than every competitor, Auglab included. Per task it has the lowest HD95 on BraTS-GLI, CHAOS, ON-Harmony, Open-MS, Spine and Pelvis (on CHAOS the gap to SRCSM is not significant); it ties Auglab on Breast without a significant gap; and on Mandible it is significantly

Setting	Base	noEM	EM	SRCSM	Auglab	Ours
BraTS-GLI T1n	25.3*	8.7*	43.1*	40.3*	46.3	46.4
BraTS-GLI T1c	28.3*	9.5*	42.9*	44.5*	46.6	47.1
BraTS-GLI T2w	18.1*	8.4*	44.1*	43.8*	47.1	47.3
BraTS-GLI FLAIR	21.8*	8.7*	42.3	37.5*	39.4*	43.0
CHAOS T1in	29.5*	63.3*	84.4*	84.5	84.7*	86.0
CHAOS T2spir	29.6*	58.5*	83.5*	84.8*	84.5*	85.8
ON-Harmony T1w	27.5*	66.0*	67.3*	66.4	67.6*	68.6
ON-Harmony T2w	34.8*	66.7	67.5	66.7	67.2	67.7
ON-Harmony DWI	42.6*	61.6	64.2	66.8	63.4	63.1
Open-MS FLAIR	25.2*	4.6*	32.8*	34.6*	32.8*	39.5
Open-MS T1w	16.7*	4.0*	35.1*	31.4*	35.8	37.4
Mandible CBCT	59.2*	53.6*	76.3*	79.0	78.3	77.7
Breast T1-CE	34.1*	16.6*	40.4*	18.0*	41.1*	41.9
Breast T2w	35.9*	16.5*	38.9*	14.8*	39.0*	40.5
Spine CT	27.0*	46.7*	79.1*	—	77.5*	80.2
Spine Dixon in	69.9*	48.4*	82.2*	—	84.2	84.3
Pelvis CT	35.6*	17.1*	66.5*	71.4	68.5	69.2
Pelvis MRI	24.0*	12.8*	58.6*	59.2*	60.0	62.0

Table 5. Per-setting Dice (%), best per row in bold. Each row is one training modality of one task, scored on exactly the held-out contrasts and external cohorts of its task column in Tab. 1, so the rows of a task average to that column. *: PALETTE-Aug significantly better than that method within the setting (same patient-level sign-flip test, Holm-corrected across competitors). Spine has no SRCSM run.

Setting	Base	noEM	EM	SRCSM	Auglab	Ours
BraTS-GLI T1n	41.0*	78.0*	18.4*	20.8*	15.1	15.4
BraTS-GLI T1c	40.3*	78.3*	19.5*	14.9	15.1	15.4
BraTS-GLI T2w	54.9*	79.7*	16.5*	15.7*	14.8	15.0
BraTS-GLI FLAIR	43.1*	76.0*	17.3	18.5*	19.2*	16.6
CHAOS T1in	119.0*	78.0*	25.4	22.7	30.4*	22.3
CHAOS T2spir	125.1*	83.2*	22.0	26.0	25.1	23.2
ON-Harmony T1w	33.4*	10.4*	5.4	6.7	5.2*	4.3
ON-Harmony T2w	23.7*	4.6*	4.0	4.6*	4.2*	4.0
ON-Harmony DWI	14.7*	4.9	4.6	4.5	4.6	4.7
Open-MS FLAIR	39.9*	41.5*	23.6	23.8	26.6	24.3
Open-MS T1w	30.5*	42.3*	25.9	25.5	26.3*	24.2
Mandible CBCT	21.7	32.9*	24.9*	19.3	22.1	23.5
Breast T1-CE	61.8*	103.2*	61.7*	92.4*	57.3	59.0
Breast T2w	55.2	99.1*	60.1*	91.2*	56.9	56.3
Spine CT	78.3*	110.4*	4.5*	—	5.8*	3.9
Spine Dixon in	21.6*	113.2*	3.5*	—	3.6	3.1
Pelvis CT	146.2*	252.0*	20.0	16.0	23.5	21.6
Pelvis MRI	152.3*	271.6*	48.0	51.9*	47.8*	36.4

Table 6. Per-setting HD95 (mm, ↓), best per row in bold. Each row is one training modality of one task, scored on exactly the held-out contrasts and external cohorts of its task column in Tab. 1, so the rows of a task average to that column. *: PALETTE-Aug significantly better than that method within the setting (same patient-level sign-flip test, Holm-corrected across competitors). Spine has no SRCSM run.

behind SRCSM, Auglab and the baseline, the one task where the baseline beats PALETTE-Aug. When HD95 dis-

agrees with Dice for a method – a noticeably better HD95 than its Dice would predict – the usual explanation is low recall: a method that predicts only a small, confident, well-localized fragment of the structure can have a deceptively good boundary distance while missing most of the target, so a strong HD95 paired with a weak Dice is a warning sign rather than evidence of competitiveness.

Full ladder: numbers and significance. Fig. 2 (main paper) shows the rung sequence visually; Tab. 9 gives the underlying out-of-domain Dice at every rung of all fourteen ladders, with significance for each step against the rung immediately before it (Holm-corrected paired Wilcoxon on patients, within each ladder’s family of steps). Most steps that add an ingredient help, as expected. Eight cells are significant *decreases* (marked ↓). Two are the fill-swap step on ON-Harmony T1w and T2w. Two are the Voronoi sub-parcellation step, on BraTS-GLI T1n and CHAOS T1in, which the real-fill step then more than recovers. Three are the +Auglab rung — on ON-Harmony DWI, CHAOS T1in and Mandible, the task where the complete method places third (Sec. 4.2) — and one is Mandible’s val000→val100 step. There PALETTE-Aug alone (the +real rung, 69.5) is the ladder’s maximum and adding the Auglab recipe takes it back down; adding PALETTE-Aug to Auglab also lowers Mandible Dice (Tab. 1), so the two components interact negatively there and we cannot attribute the shortfall to either. Both CHAOS ladders peak at the same rung. We have not diagnosed which Auglab operator is responsible.

We likewise do not hide that Voronoi’s own marginal contribution is task-dependent rather than uniformly positive – a separate paired significance check on all-contrast Dice (the estimand of Tab. 13, not the OOD slice above) shows: Voronoi significantly *helps* Open-MS FLAIR and (by a small margin) ON-Harmony T1w, is flat on CHAOS T1in (its OOD step above is a small significant decrease), and significantly *hurts* BraTS-GLI T1n under the real-fill deployed configuration as well, not only here under noise fill. The full method is still best on BraTS-GLI T1n (Tab. 5; not significantly ahead of Auglab): whatever Voronoi costs there is more than recovered by the subsequent real-intensity fill step, so the net effect of the complete pipeline is positive even though one internal component is not.

Per-label breakdown: is the fill-swap effect uniform within a task? BraTS-GLI T2w’s pooled fill-swap effect is small and not significant (Tab. 2), with one significantly negative held-out contrast inside it (T2w→T1n, −4.1 Dice). This raises a further question: within that one (training, eval) pair, does the fill-swap step affect every anatomical sub-region equally, or is the worsening itself concentrated in specific labels? We break BraTS-GLI T2w→T1n down

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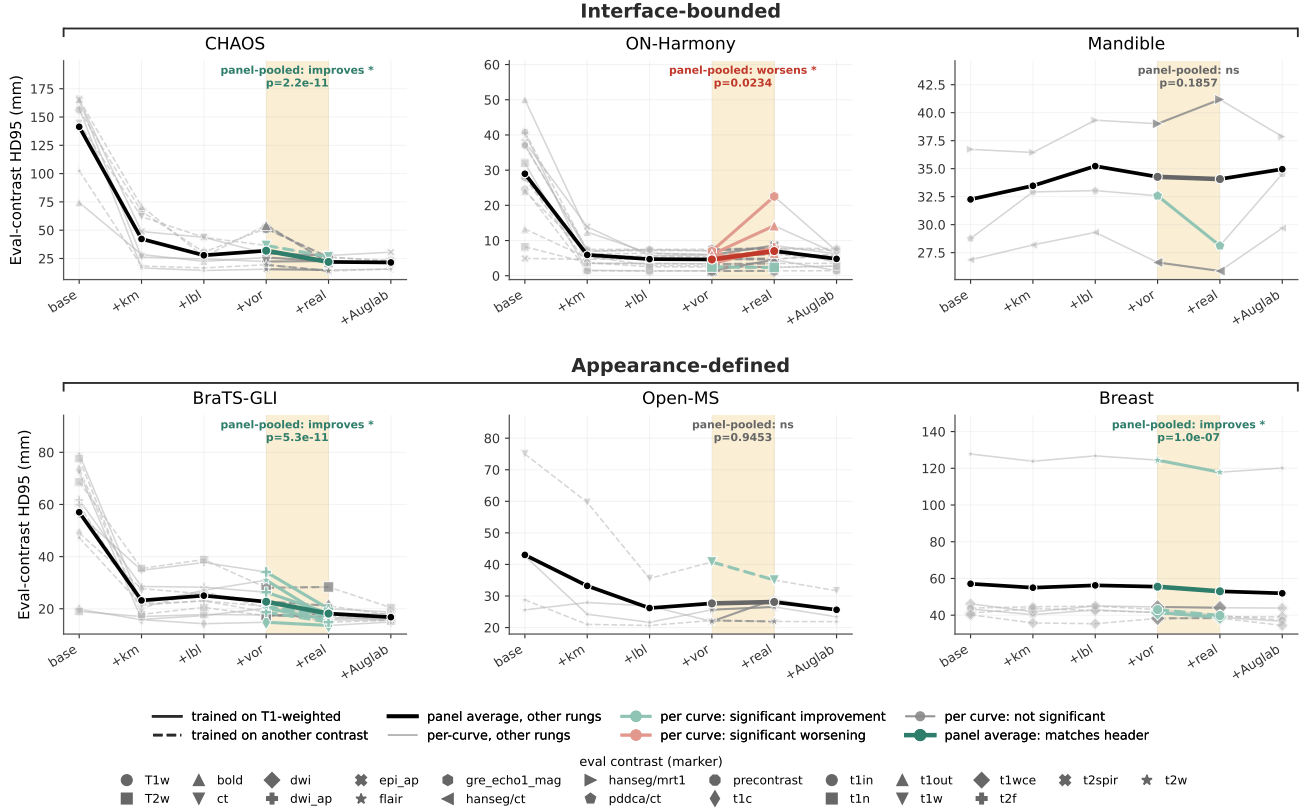
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	BraTS-GLI	CHAOS	ON-Harm.	Open-MS	Mandible	Breast	Spine	Pelvis	p vs ours
Baseline	44.8	122.1	24.0	35.2	21.7	58.5	49.9	149.3	4.7e-48
SynthSeg-noEM	78.0	80.6	6.7	41.9	32.9	101.1	111.8	261.8	2.9e-58
SynthSeg-EM	17.9	23.7	4.7	24.7	24.9	60.9	4.0	34.0	1.8e-6
SRCSM	17.5	24.4	5.3	24.6	19.3	91.8	—	33.9	2.0e-34
Auglab	16.1	27.8	4.7	26.5	22.1	57.1	4.7	35.6	1.0e-4
PALETTE-Aug (ours)	15.6	22.7	4.3	24.3	23.5	57.7	3.5	29.0	—

Table 7. Task-level HD95 (mm, ↓). Same pooling and test as Tab. 1, and likewise no cross-task average. Spine has no SRCSM run.

Figure 4. The causal-ablation ladder of Fig. 2 (main paper), same per-dataset panels, same boundary-type grouping, same fill-swap rung and same visual scheme (per-curve color only on the real-fill segment; bold black panel average; one panel-pooled p -value per panel), reported on HD95 instead of Dice (*improvement* means the fill-swap step lowers HD95). BraTS-GLI, CHAOS and Breast improve significantly at panel level and ON-Harmony worsens; Open-MS and ToothFairy2 do not move significantly. Underlying per-row deltas in Tab. 8.

by its four labels – necrotic core (NCR), oedema (SNFH), enhancing tumor (ET), resection cavity (RC); $n = 70$ test cases for SNFH and ET, present in essentially every case (NCR $n = 51$, RC $n = 67$, since not every tumor has a necrotic core or a resection cavity).

The worsening is concentrated in SNFH (oedema) (Tabs. 10 and 11). SNFH has the largest gap between directions (+19.9 Dice) and the largest significant sign reversal: +14.7 when trained on T1n, −8.8 when trained on T2w. RC reverses too, at a smaller size. NCR shows no signifi-

cant effect in either direction, and ET improves only in the T1n→T2w direction. Fig. 5 shows this on one case: with real fill, the T2w-trained model’s SNFH prediction shrinks to a thin band near the ventricle, while its ET prediction is almost unchanged.

One candidate mechanism, flagged for future work rather than tested here: a texture pattern learned from one contrast’s rendering of a diffuse, texture-defined label like oedema may not simply fail to transfer to a different contrast’s rendering of the same tissue – it may actively mis-

Task	target type	Δ HD95 (mm)	p
CHAOS T1in	interface-bounded	-1.36	0.80
CHAOS T2spir	interface-bounded	-18.47	3.4e-12
ON-Harmony T1w	interface-bounded	+6.82	0.062
ON-Harmony T2w	interface-bounded	+0.38	0.062
ON-Harmony DWI	interface-bounded	-0.05	0.75
Mandible CBCT	interface-bounded	-0.20	0.37
Open-MS FLAIR	appearance	-3.02	0.14
Open-MS T1w	appearance	+3.94	0.80
BraTS-GLI T1n	appearance	-4.59	6.3e-10
BraTS-GLI T1c	appearance	-9.91	4.2e-11
BraTS-GLI T2w	appearance	-1.95	0.013
BraTS-GLI FLAIR	appearance	-1.71	0.21
Breast T1-CE	appearance	-3.21	8.8e-5
Breast T2w	appearance	-1.82	0.0024

Table 8. Same fill-swap step as Tab. 2 (main paper), on HD95 (negative is better), with p Holm-corrected across the fourteen rows. Five of the six significant improvements are on appearance-defined targets, including both Breast rows, whose Dice does not move; the sixth, and the largest in millimetres, is CHAOS T2spir. HD95 is less consistent than Dice within a task (its sign flips between the two modalities of Open-MS), so we state the claim on Dice and report HD95 here rather than as confirmation of it.

Rung	Open-MS FLAIR	Open-MS T1w	BraTS-GLI T1n	BraTS-GLI T1c	BraTS-GLI T2w	BraTS-GLI FLAIR	CHAOS T1in
base	2.4	1.1	16.5	15.9	6.0	11.0	17.0
+km	11.2*	9.8*	38.4***	29.6***	38.6***	26.1***	82.2***
+lbl	17.5*	22.4*	41.8***	30.3	40.2**	26.4	86.1***
+vor	19.6	34.1*	39.8***↓	30.5	41.0	31.9***	85.7**↓
+real	27.3*	34.9	47.0***	43.7***	41.8	31.7	87.1***
+AL(v000)	30.6	36.4	46.9	44.2	46.8***	40.9***	86.4***↓
+AL(v100)	31.3	34.7	47.6*	46.5***	46.9	41.2	86.1

Rung	CHAOS T2spir	ON-Harm. T1w	ON-Harm. T2w	ON-Harm. DWI	Mandible CBCT	Breast T1-CE	Breast T2w
base	19.7	14.8	24.5	36.1	39.4	23.7	33.5
+km	77.4***	62.5*	63.5*	60.4*	64.6***	28.8***	37.0***
+lbl	81.4***	62.9	63.6	64.5*	65.6***	28.5	38.8***
+vor	82.0***	63.2	64.2*	64.6	68.1***	30.8***	39.8**
+real	87.1***	58.0*↓	62.6*↓	65.4*	69.5***	29.9	40.3*
+AL(v000)	85.4	64.2	63.9	60.9*↓	67.6***↓	34.7***	41.9***
+AL(v100)	85.9	64.2	64.4*	60.9	67.5***↓	—	—

Table 9. Out-of-domain Dice (%) at every rung of every ladder (Fig. 2), with significance for each step against the rung above it (val100 against val000): * $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$ (Holm-corrected paired Wilcoxon on patients, within each ladder); ↓ marks a significant decrease. Rungs: base(line), + k -means, +label-remap, +Voronoi (noise fill), +real-intensity fill (PALETTE-Aug alone), +Auglab (val000/val100). Open-MS and ON-Harmony have eight test units each, so their smallest attainable p after correction is just under 0.05. I-SPY2 has no val100 arm. Mandible’s held-out axis is cross-dataset.

lead, if the model has come to rely on a contrast-specific cue that the other contrast does not reproduce. This would be an instance of shortcut learning [14] – specifically, a network relying on a domain-specific cue that does not reproduce across acquisition protocols, the mechanism Ouyang et al.

Label	T1n→T2w	T2w→T1n	gap
NCR	21.5	20.0	+1.4
SNFH	66.3	46.4	+19.9
ET	27.5	29.0	-1.5
RC	52.3	42.8	+9.5

Table 10. Absolute OOD Dice (%) of the real-fill rung (PALETTE-Aug alone), per label, both transfer directions.

Label	T1n→T2w Δ	T2w→T1n Δ
NCR	+1.9 ($p=0.10$)	+1.0 ($p=0.77$)
SNFH	+14.7 ($p=4.3e-11$)	-8.8 ($p=1e-4$)
ET	+2.3 ($p=0.017$)	+1.6 ($p=0.24$)
RC	+2.9 ($p=0.0003$)	-6.9 ($p=0.0004$)

Table 11. Fill-swap step (+Voronoi noise fill → real fill), per label, both directions. p : Holm-corrected paired Wilcoxon, family = the 4 labels within each direction. **Bold** = significant.

BraTSGLI02108101, fold 0 → T2w-trained model evaluated on T1n

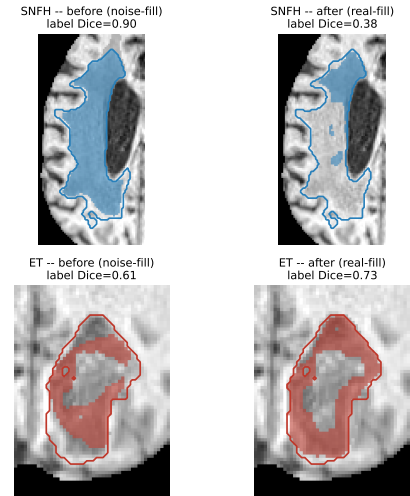


Figure 5. BraTSGLI02108101, fold 0, T2w-trained model evaluated on T1n. Zoomed crops before (noise-fill) and after (real-fill) the fill-swap step, for the SNFH-dominant (top) and ET-dominant (bottom) lesion components of the same case. Color = ground truth wherever the prediction agrees; grey = false positive; outline = ground truth.

[36] identify behind failed single-source domain generalization in cross-modality and cross-sequence medical segmentation – rather than the intended, causal feature, which a simplicity-biased network need not prefer even when it would generalize better [43]. We have not tested this mechanistically here; it is the most concrete hypothesis consistent with the label-specific, direction-dependent pattern above, not a demonstrated cause.

Task	Dice $\uparrow$		HD95 (mm) $\downarrow$	
	val000	val100	val000	val100
BraTS-GLI	45.9	46.6	15.6	15.1
CHAOS	85.9	86.0	22.7	20.6
ON-Harmony	66.5	66.6	4.3	4.3
Open-MS	38.5	37.9	24.3	24.2

Table 12. Applying PALETTE-Aug at validation time as well as at training time, on the four tasks with a task-level table for both configurations (Mandible’s val100 arm is scored only in its ladder, Tab. 9; the Breast, Spine and Pelvis arms were not trained). Across these tasks the two show no significant difference in either direction (same cross-task test as Tab. 1; two-sided Dice $p = 0.74$, HD95 $p = 0.096$), though val100 is significantly ahead within BraTS-GLI and ON-Harmony individually. We report the like-for-like val000 configuration throughout the main paper.

10. Synthetic contrast at validation time

Throughout the main paper PALETTE-Aug is applied with synthetic contrast during training only (*train050/val000*), because every competing method is likewise evaluated with a real-contrast validation set; this keeps the comparison exactly like-for-like. PALETTE-Aug can additionally be applied to the validation set, so that model selection is performed under randomized contrast as well (*train050/val100*).

Validation-time randomization improves most tasks on both metrics (Tab. 12), most visibly on boundary distance for CHAOS (22.7 $\rightarrow$ 20.6 mm). The effect is not significant across tasks, and Open-MS Dice moves slightly the other way, so we do not claim it as a separate contribution; we report it because it costs nothing and because it indicates that the benefit of contrast randomization extends to model selection and is not confined to the gradient updates.

11. Component ablation

To attribute the gain of the full method to its individual components, rather than to synthetic-contrast augmentation in general, we build an additive ablation ladder on top of the nnU-Net baseline (*Base*, Sec. 4.1): first the unsupervised k -means intensity partition alone (§3.1, step 1); then the anatomical label-remap step (step 4); then the Voronoi spatial sub-parcellation (step 2); and finally the full signed affine remap of Eq. (1) (step 3), which together constitute PALETTE-Aug on its own. Every rung is evaluated without Auglab, so that the ladder isolates PALETTE-Aug’s own internal components from the separate question of whether it composes with a prior augmentation pipeline, which we address next. The synthetic-data fraction is fixed at 50% of training iterations throughout, as one of several augmentation constants (alongside, e.g., the k -means class count and the affine magnitude range of §3.1) that are set once and not

Component	CHAOS T1in	Open-MS FLAIR	BraTS-GLI T1n	C
Base (nnU-Net default)	36.1	25.2	26.0	
+ k -means partition	84.8	30.6	41.7	
+ label remap	87.7	34.6	44.3	
+ Voronoi sub-parcellation	87.2	36.1	42.8	
+ affine remap (PALETTE-Aug alone)	87.9	40.4	48.5	
+ Auglab (Ours , deployed)	87.4	39.5	47.5	

Table 13. Mechanism ablation (top five rows, additive, no Auglab) and deployment composability check (bottom row). Each rung adds exactly one PALETTE-Aug component on top of the row above. Dice (%) over all test contrasts of the setting, the training contrast included, with equal weight per contrast (the all-contrast series of the same ladders as Tab. 9, which reports the held-out contrasts only). CHAOS T1in: CHAOS’s own test contrasts only, without the external cohorts.

tuned per dataset; validation uses real-contrast data only (no synthetic contrast at validation time).

Composability with Auglab. PALETTE-Aug is deployed, throughout this paper, as one additional operator inside the Auglab pipeline rather than as a replacement for it (Sec. 4.1); Auglab is a full prior augmentation method in its own right, not a component of PALETTE-Aug, so stacking it on top of the ladder above is a composability/deployment check, not a further mechanism-ablation step. Tab. 13 keeps the two separate: the top five rows isolate PALETTE-Aug’s internal mechanism, evaluated with no Auglab throughout, while the bottom row reports PALETTE-Aug plus Auglab — the configuration used everywhere else in this paper — set apart by a rule rather than presented as the next rung of the same ladder.

12. Texture fidelity per method

The noise-versus-real NGF analysis is in the main paper (Fig. 3); Tab. 14 gives the per-method values it cites.

Why Auglab is the closest competitor. Auglab is the strongest competitor on the cross-task test, and NGF suggests why: it keeps most of the source texture, only modestly less than PALETTE-Aug and far more than the label-style methods (Tab. 14). The large margins appear against the methods that discard texture.

Method	FLAIR	T1w
PALETTE-Aug	0.809	0.894
Auglab	0.768	0.777
SynthSeg-EM	0.462	0.422
noise fill	0.457	0.401
SynthSeg-noEM	0.330	0.319

Table 14. Texture fidelity: NGF similarity to the source scan (mean over $n=30$ Open-MS scans; $\approx 1/3$ = isotropic-noise floor). The PALETTE-Aug row is the transform alone (the real-fill rung), without the Auglab pipeline.